DATE: 14/02/2026

EX-04
Analyze the Malicious Network Traffic using Wireshark

AIM:

To perform deep packet inspection on a real-world PCAP capture file using Wireshark, identifying malicious activity including malware downloads, Cobalt Strike C2 communication, post-infection traffic, malspam activity, and suspicious DNS queries in the TryHackMe Carnage room.

PROCEDURE:

- Access the TryHackMe platform and open the Carnage room.
- Open the provided PCAP file in Wireshark for analysis.
- Apply display filters (http, dns, smtp, tcp) to isolate specific traffic types.
- Examine HTTP traffic to identify the first connection to a malicious IP and timestamp.
- Inspect downloaded files by following TCP/HTTP streams to identify malicious ZIP archives.
- Identify the hosting domain, web server software, and version of the malicious file server.
- Use Wireshark's Export Objects feature to list files inside ZIP archives without downloading.
- Correlate source IPs with VirusTotal Community tab to confirm Cobalt Strike C2 servers.
- Examine DNS queries to trace post-infection domains and IP-check API calls.
- Analyse SMTP traffic to identify malspam activity, sender addresses, and packet counts.
- Document all findings including malicious domains, IPs, and communication patterns.

TASK-1

Task-2

Task-2

Answer the questions below

What was the date and time for the first HTTP connection to the malicious IP?

(**answer format**: yyyy-mm-dd hh:mm:ss)

2021-09-24 16:44:38 — ✓ Correct Answer

What is the name of the zip file that was downloaded?

documents.zip — ✓ Correct Answer

What was the domain hosting the malicious zip file?

attirenepal.com — ✓ Correct Answer

Without downloading the file, what is the name of the file in the zip file?

chart-1530076591.xls — ✓ Correct Answer

What is the name of the webserver of the malicious IP from which the zip file was downloaded?

LiteSpeed — ✓ Correct Answer

What is the version of the webserver from the previous question?

PHP/7.2.34 — ✓ Correct Answer

Malicious files were downloaded to the victim host from multiple domains. What were the three domains involved with this activity?

finejewels.com.au, thielbiagt.com, new.americold.com — ✓ Correct Answer

What are the two IP addresses of the Cobalt Strike servers? Use VirusTotal (the Community tab) to confirm if IPs are identified as Cobalt Strike C2 servers. (answer format: enter the IP

185.106.96.158, 185.125.204.174 — ✓ Correct Answer

What is the Host header for the first Cobalt Strike IP address from the previous question?

ocsp.verisign.com — ✓ Correct Answer

What is the domain name for the first IP address of the Cobalt Strike server? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

survmeter.live — ✓ Correct Answer

What is the domain name of the second Cobalt Strike server IP? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

securitybusinpuff.com — ✓ Correct Answer

What is the domain name of the post-infection traffic?

maldivehost.net — ✓ Correct Answer

What are the first eleven characters that the victim host sends out to the malicious domain involved in the post-infection traffic?

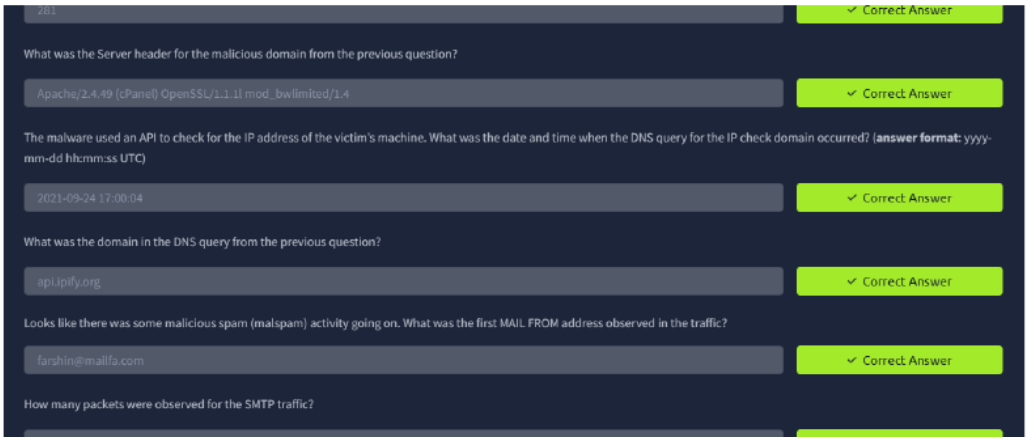
zLIisQRWZI9 — ✓ Correct Answer

What was the length for the first packet sent out to the C2 server?

281 — ✓ Correct Answer

What was the Server header for the malicious domain from the previous question?

Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4 — ✓ Correct Answer



RESULT:

The Wireshark PCAP analysis was completed successfully. Malicious HTTP traffic was identified originating on 2021-09-24, revealing the download of a ZIP-based malware payload (documents.zip) from a compromised domain. Two Cobalt Strike C2 servers (185.106.96.158 and 185.125.204.174) were confirmed via VirusTotal. Post-infection traffic to maldivehost.net was traced, alongside DNS-based IP-check queries to api.ipify.org. Malspam activity was detected in 1439 SMTP packets, with the first sender address identified as farhin@mailfa.com — confirming a fully compromised system conducting multi-stage attacker activity.